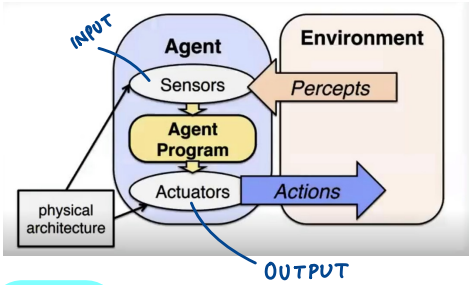
Sesi 1 & 2

Agent → anything that can be viewed as perceiving its environment through sensors & acting upon that environment through actuators.



PEAS

What is PEAS for? u/ define spesifikasi rational agent

- Autonomous car


[source]

– Performance: safety, destination, legality, comfort → goalnya apa?
– Environment: streets, pedestrian, highway, weather → tempat agent kerja/ lingkungan yg pengaruh
– Actuators: steering, accelerator, brake, horn → apa yg bs dilakuin?
– Sensors: video, GPS, accelerometer, keyboard → gmn ngumpulin info?

Agent Types types of Intelligent Agents

1. **Simple Reflex Agents**
   sederhana, cepat, gabisa untuk case kompleks
   Cth. automatic lights

2. **Model-based Reflex Agents**
   smarter, more complex, bisa tau history-nya / belajar dari pngalamn
   Cth. robot vacum navigation, Google Maps

3. **Goal-based Agents** fokus ke tujuannya
   kombinasi model sama goal informationnya, searching & planning dulu
   Cth. delivery drone dari Amazon

4. **Utility-based Agents**
   based on agent happiness (measure utility).
   Cth. Netflix's recommendation system, dari faktor ratings, preferences

5. **Learning-based Agents**
   belajar dari feedback, knowledge, changes, learning goals

Sesi 3 & 4

Cth: Chat Bot, Chat GPT

REPRESENTING A PROBLEM: THE FORMALISM
SOLUTION

- Search tree: models the sequence of actions
  - Root: initial state
  - Branches: actions
  - Nodes: results from actions
  - Expanding: Process of generating child nodes (performing an action)

REPRESENTING A PROBLEM: THE FORMALISM
STATE SPACE

- To formulate a problem as a search problem we need the following components:
  - The initial state : The state where the agent starts
  - Actions : Possible actions available to the agents
  - Transition model : A description what an action does
  - Goal test : Check if a given state is a goal state
  - Path cost : Measure the cost to achieve the goal
    → jumlah jarak / waktu buat ke goals

REPRESENTING A PROBLEM: THE FORMALISM
EXAMPLE: ROMANIA TRAVEL

- State space:
  - ❑ The initial state : In (Arad)
  - ❑ Actions : Go (Zerind), Go (Zibiu), etc.
  - ❑ Transition model : Results (In (Arad), Go (Zerind)) = In(Zerind)
  - ❑ Goal test : In (Bucharest)
  - ❑ Path cost : Path length in kilometers

REPRESENTING A PROBLEM: THE FORMALISM
SEARCHING SOLUTION

- Measuring problem-solving performance
  - ➢ Completeness: Is the algorithm guaranteed to find a solution when there is one?
  - ➢ Optimality: Does the strategy find the optimal solution?
  - ➢ Time complexity: How long does it take to find a solution?
  - ➢ Space complexity: How much memory is needed to perform the search?

Search Strategies

1) Uniformed Search (gak ada heuristics) / Blinded Search

1 Breath-First Search (BFS) FIFO
↳ baca dari kiri ke kanan kayak Queue. Baris per baris.
(+) Complete : pasti ketemu karena search menyeluruh ke semua.
(−) Time complexity : lama / lambat karena perlu cari satu"
(−) Space complexity : perlu banyak memori karena jadi ingat banyak

2 Depth-First Search (DFS) LIFO
↳ Stack. dari root atas ke leaf dulu satu-satu ke bawah baru ke samping
(+) Time complexity : Cepat
(−) Complete : kalau cabang / leaf banyak, bisa ga selesai"

3 Depth-Limited Search
↳ sama kayak DFS cuma dikasih batasan aja kalau sampe branch ke-sekian belum ketemu, maka next ke yang lain.
(+) Time complexity : Cepat
(−) Complete : bisa ga selesai

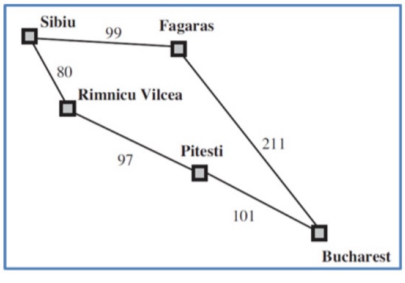
4 Iterative Deepening Search (IDS) harusnya gak masuk ujian, kepanjangan
↳ kombinasi DFS dan BFS
(+) Completeness : pasti ketemu & selesai
(−) Time complexity : makan banyak waktu dan biaya u/ dikerjakan

5 Uniform-cost Search (UCS) g(n)
↳ expand node dengan jarak paling kecil. The lower path cost g(n)
(+) Optimality : strategi find optimal solution
(−) Time complexity : lama / lambat karena perlu cari satu"

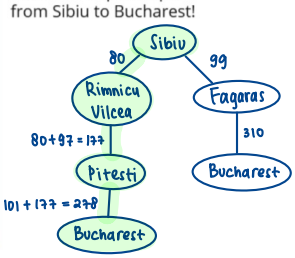
paling keluar

Contoh.

UNINFORMED SEARCH
UNIFORM COST SEARCH: EXAMPLE

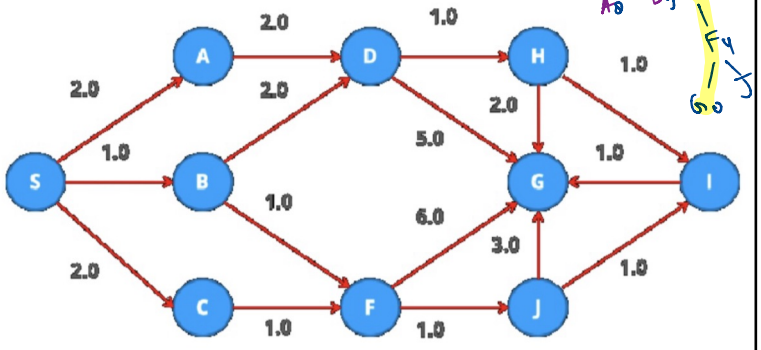


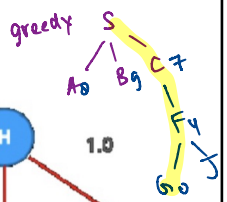
Search the optimal path from Sibiu to Bucharest!



Latihan Soal VN → visited note

CASE STUDY
SIMULATING SEARCHING





BFS

FIFO Queue : S, A, B, C, D, F, H, G, J, I

VN : S, A, B, C, D, F, H, G

216 246

root → parent → child

DFS

LIFO Stack :

	Stack	VN
1 of child	G	S
	H	A
	D	D
parent	A	H
	B	G
	C	
root	S	

atau

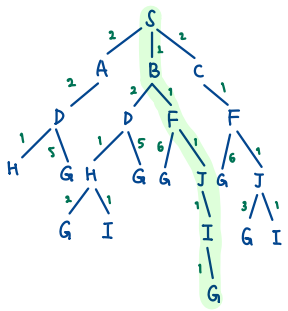
VN
S
C
F
J
G

tergantung urutan stack

root → child

UCS

g(n) = S, B, F, J, I, G



2) Informed Search (ada nilai heuristics)

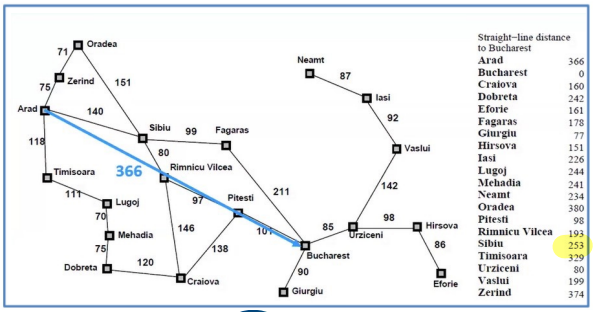
1 (Greedy) Best-First Search (BFS) h(n)

(+) Time complexity : Cepat

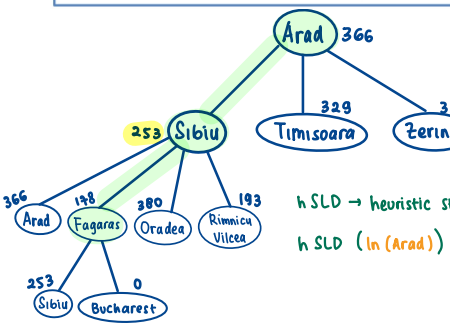
(-) Optimality : kurang optimal

Contoh.

INFORMED SEARCH
GREEDY BEST FIRST SEARCH



Straight-line distance to Bucharest	
Arad	366
Bucharest	0
Craiova	160
Dobreta	242
Eforie	161
Fagaras	178
Giurgiu	77
Hirsova	151
Iasi	226
Lugoj	244
Mehadia	241
Neamt	234
Oradea	380
Pitesti	98
Rimnicu Vilcea	193
Sibiu	253
Timisoara	329
Urziceni	80
Vaslui	199
Zerind	374



h SLD → heuristic straight line distance

h SLD (In (Arad)) = 366

2 A* (A star) f(n) = g(n) + h(n)

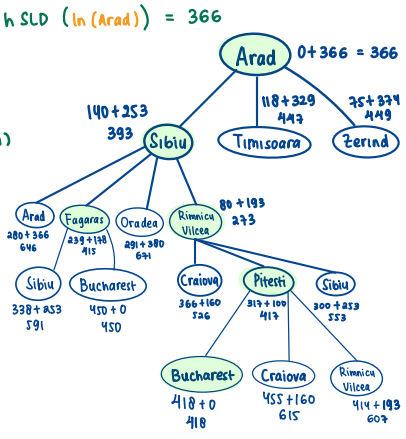
↳ gabungan UCS & greedy

(+) Time complexity : Cepat

(+) Optimality :

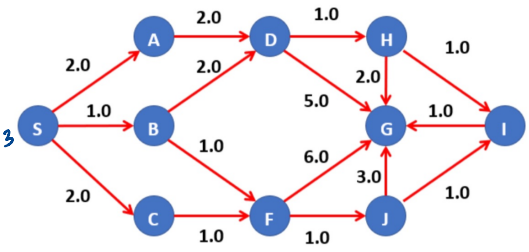
Strategi find optimal solution

(-) Space complexity : perlu banyak memori



Latihan Soal.

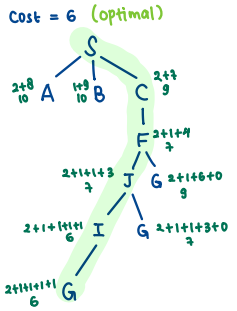
Searching Problem



- Using A*

S-C-F-J-I-G

Cost = 6 (optimal)



Heuristic Function	
h(A)	8.0
h(B)	9.0
h(C)	7.0
h(D)	5.0
h(F)	4.0
h(G)	0
h(H)	2.0
h(I)	1.0
h(J)	3.0

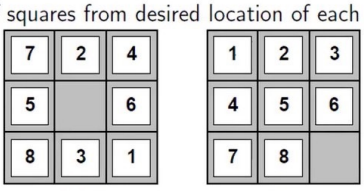
- **Greedy** lebih cepat tetapi tidak optimal (tidak mempertimbangkan cost).
- **A*** lebih lambat tetapi **menemukan jalur dengan cost terendah.**

Question: simulate the node expansion from start state (S) to reach goal state (G) using:
- Breadth First Search
- Depth First Search
- Uniform Cost Search
- Greedy → S → C → F → G , cost = 9 (tdk optimal)
- A*

Heuristic Functions

E.g., for the 8-puzzle:

$h_1(n)$ = number of misplaced tiles
$h_2(n)$ = total Manhattan distance
(i.e., no. of squares from desired location of each tile)

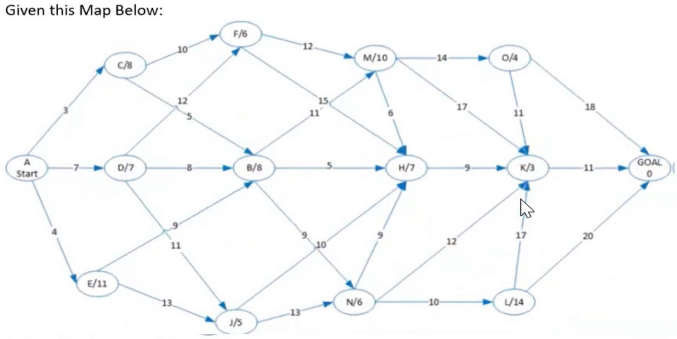


$h_1(S)$ =?? 6

$h_2(S)$ =?? 4+0+3+3+1+0+2+1 = 14

Latihan Soal dari soal UTS tahun lalu.

1. Given this Map Below:



1. Simulate the traversal Process using:
   a. BFS
   b. DFS
2. Find the optimal tour (optimal path) and write the length of the path for graph Travelling Salesman Problem below, using:
   a. Best First Search (Greedy) Algorithm
   b. A-star Algorithm

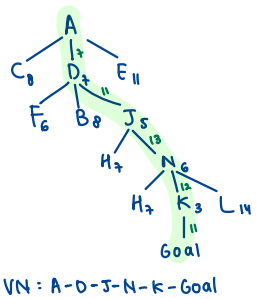
1. a) Queue : A, C, D, E, F, B, J, M, H, N, O, K, L, Goal

VN : A, C, D, E, F, B, J, M, H, N, O, K, L, Goal

b) Stack : ~~A~~ E D ~~C~~ B ~~F~~ H ~~M~~ K ~~O~~ ~~Goal~~

VN : A, C, F, M, O, Goal

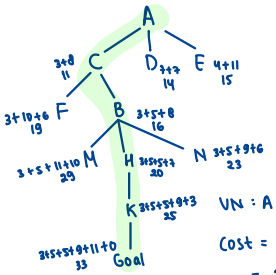
2. a) greedy



VN : A-D-J-N-K-Goal

Cost = 54

b) A*

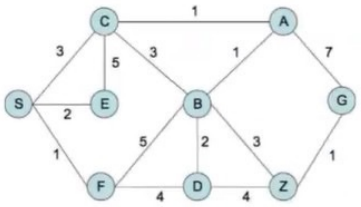


VN : A-C-B-H-K-Goal

Cost = 3 + 5 + 5 + 9 + 11

= 33

2. Given This Map:



Node	h(n)
S	8
F	6
E	6
C	5
D	4
B	3
A	2
Z	1
G	0

1. Simulate the traversal Process using:
   a. BFS
   b. DFS
2. Find the optimal tour (optimal path) from S to G and write the length of the path for graph Travelling Salesman Problem below, using:
   a. Best First Search (Greedy) Algorithm
   b. A-star Algorithm

a) BFS

ZIG ZAG

root → parent → child

node : S, C, E, F, A, B, D, G, Z

UN : S, C, E, F, A, B, D, G

b) DFS

Stack :



1 of child

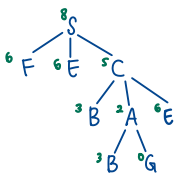
parent

root

root → child

UN : S - C - A - G

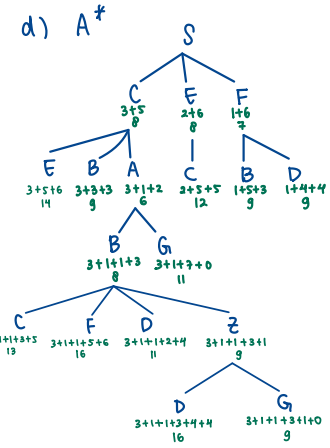
c) Greedy



Node = S, C, A, G

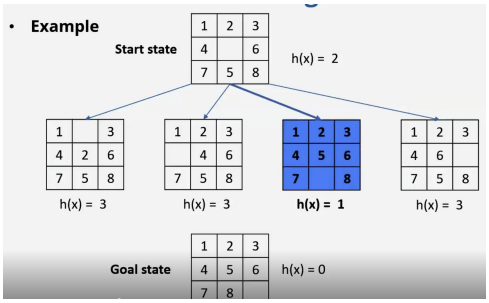
h(n) = 11

d) A*

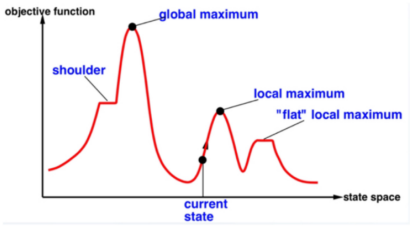


Sesi 5 - Local Search

↳ u/ nyelesain masalah yg lebih real-world. Urutan gk penting



Kurva yg menggambarkan hasil percobaan :



Cek atasnya	
max	min
B < V = ?	α > V = ?
C = 8 > -∞	8 < ∞ = C
C = 10 > 4	3 < 8 = C
P = 0 < 4	7 < ∞ = C
C = 13 > 4	-5 < 10 = C
C = 100 > 4	1 < 13 = C
	150 > 100 = P

1) Hill Climbing Search (Steepest - Ascent)

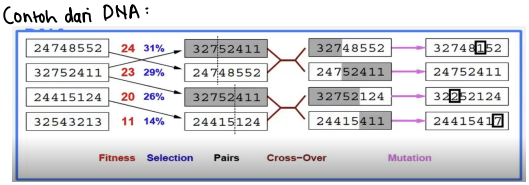
- Ridges : ada banyak local maximum
- Plateu : datar terus . tidak naik / tidak turun

Solusi → pilih angka random (pendekatan stokastik) mengharapkan hasil yang bagus.

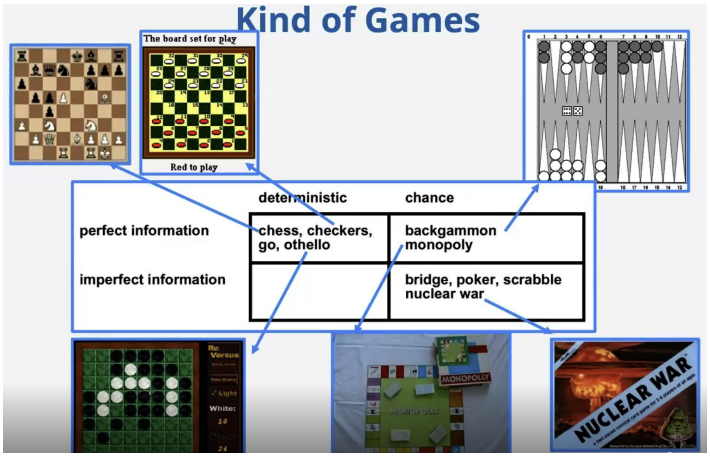
2) Genetic Algorithm

↳ successor state-nya at least diturunkan dari 2 atau lebih parent.

Contoh dari DNA :

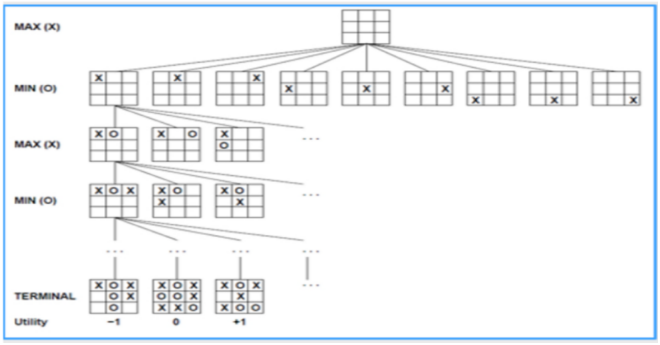


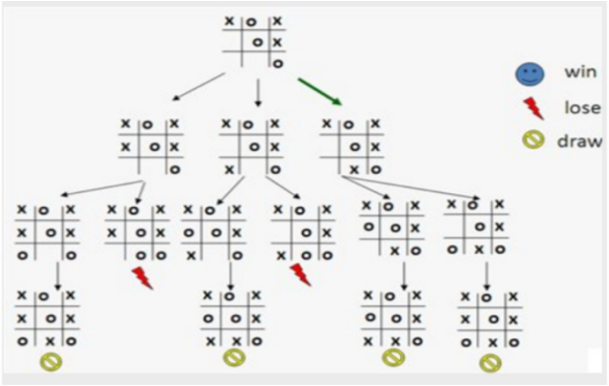
Sesi 6 - Adversial Search



Minimax Strategy

1) Game tree

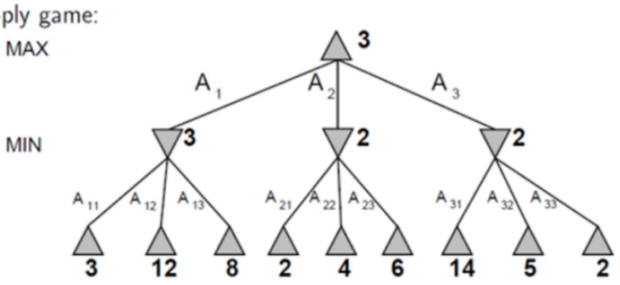




Perfect play for deterministic, perfect-information games

Idea: choose move to position with highest minimax value
= best achievable payoff against best play

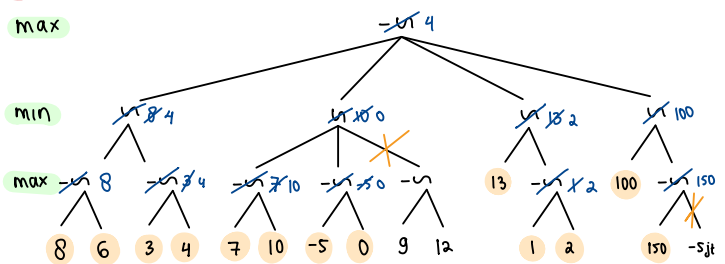
E.g., 2-ply game:



2) α - β prunning

min max

minimax simulator !!



Quiz

1. Intelligent Agent

Answer the following questions:

a. Explain 4 types of Intelligent agents and give examples!

b. An autonomous car uses various sensors and AI algorithms to navigate the road.

i. Identify the type of intelligent agent used in an autonomous car and explain how the agent works in decision-making while driving.

ii. What is the difference between a Model-Based Reflex Agent and a Goal-Based Agent in an autonomous car system?

1. a) 4 types of Intelligent Agent :

1. Simple Reflex Agents

agent yang paling sederhana, cepat, gabisa untuk case kompleks. Hanya bekerja pada kondisi saat itu aja.

Cth: automatic lights yang sensor gerak, sensor automatic buka tutup pintu.

2. Model-based Reflex Agents

agent yang smarter, more complex, bisa tau history-nya / belajar dari pngalaman. Cth: robot vacum navigation

3. Goal-based Agents

agent yang bekerja berdasarkan tujuan. Kombinasi model sama goal informationnya, akan searching & planning dulu.

Cth: delivery drone dari Amazon

4. Utility-based Agents

capai tujuan based on agent happiness (measure utility).

Cth. Netflix's recommendation system, dari faktor ratings, preferences.

5. Learning-based Agents

agent yang terus belajar dari feedback, knowledge, changes, learning goals. Contoh: Chatbot seperti Chat GPT

b) (i) Intelligent Agent used in autonomous car

Autonomous car bisa menggunakan :

⇒ Model-based Reflex Agents

⟶ bisa cepat tanggap untuk rem saat ada halangan

⇒ Goal-based Reflex Agents

⟶ bisa kasih jalur yang optimal untuk sampai ke tujuan

⇒ learning-based Reflex Agents

⟶ bisa memilih jalur yang sudah sering di-prefer driver.

Agents work process in decision making while driving

dengan menggunakan berbagai sensors

Performance : safety, destination, legality, comfort

Environment : jalanan, cuaca, pejalan kaki, lampu merah, mobil sekitar

Actuators : steering, gas, rem, klakson

Sensors : video, GPS, accelerometer

Caranya :

1. System mengumpulkan informasi data yang diperlukan, seperti mobil di sekitar, rute jalan, dll.

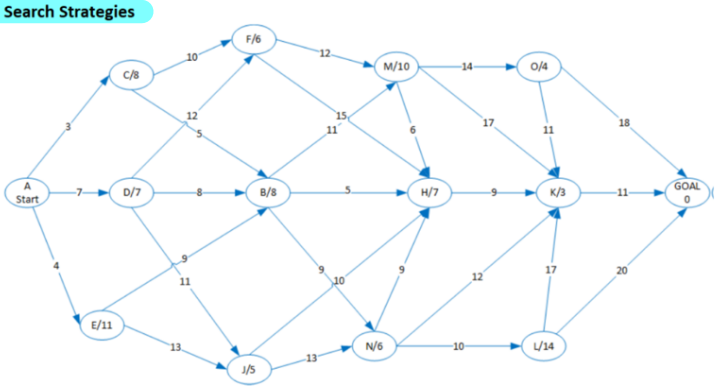
2. memilih / memutuskan cara yang paling optimal berdasarkan data yang dimiliki

3. Evaluasi, terus belajar dan improving dari data yang sudah terjadi.

(ii)

	Autonomous Car	
	Model-Based Reflex Agents	Goal-Based Reflex Agents
pengertian	membuat keputusan instan	menentukan cara yang paling optimal untuk peroleh tujuan
contoh	mobil bisa langsung rem kalau ada pejalan kaki / kendaraan lain	system memilih rute tercepat & teraman untuk sampai tujuan dengan mempertimbangkan traffic, cuaca, dll

2. Search Strategies



Find the optimal tour (optimal path) and write the length of the path for graph Travelling Salesman Problem below, using:

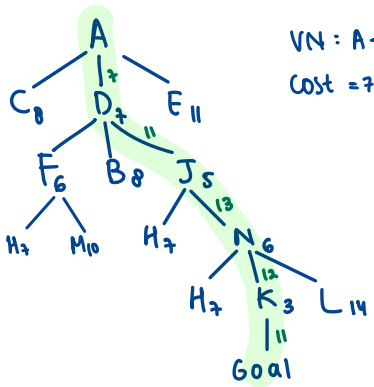
a. UCS ⟶ di halaman belakang

b. Best First Search (Greedy) Algorithm

c. A-star Algorithm

d. Explain which algorithm performs better in terms of time complexity and the best solution provided? ⟶ di hal belakang

2. b) greedy

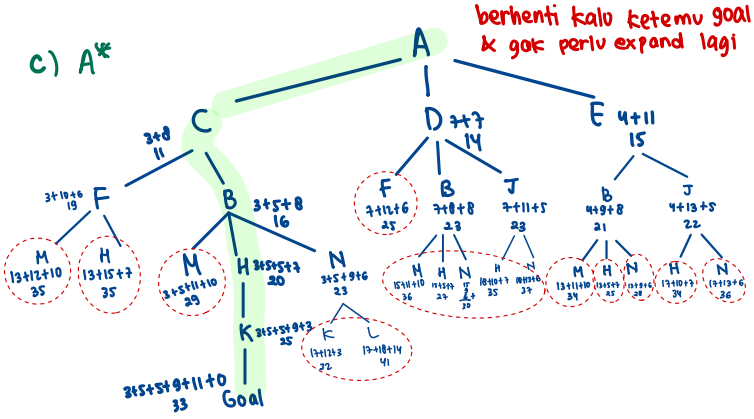


VN : A-D-J-N-K-Goal

Cost = 7+11+13+12+11 = 54

c) A*

berhenti kalo ketemu goal & gak perlu expand lagi



VN : A-C-B-H-K-Goal

Cost = 3+5+5+9+11 = 33